Homework 4
APPM 4490/5490 Theory of Machine Learning, Spring 2026

Due date: Wednesday, Feb 11 '26, before noon, via paper or via Gradescope

Instructor: Prof. Becker
Revision date: 2/9/2026

Theme: Covering numbers and Rademacher complexity-like stuff

Instructions Collaboration with your fellow students is OK and in fact recommended, although direct copying is not allowed. The internet is allowed for basic tasks (e.g., looking up definitions on wikipedia) but it is not permissible to search for proofs or to *post* requests for help on forums such as `http://math.stackexchange.com/` or to look at solution manuals; see also the class AI policy on our website. Please write down the names of the students that you worked with.

An arbitrary subset of these questions will be graded.

Reading You are responsible for reading chapter 6 of "Understanding Machine Learning" by Shai Shalev-Shwartz and Shai Ben-David (2014, Cambridge University Press). Note: you can buy the book for about $45 on Amazon (or less for an e-book), and the authors host a free PDF copy on their website (but note that this PDF has different page numbers).

Problem 1: Let $N(A,\epsilon)$ be the covering number of the set $A \in \mathbb{R}^d$ with respect to the Euclidean norm. To be precise, we define an ϵ-net of A to be a set $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_m\} \subseteq A$ such that $A \subseteq \bigcup_{i=1}^n B_\epsilon(\boldsymbol{x}_i)$ where $B_\epsilon(\boldsymbol{x}) = \{\boldsymbol{y} : \|\boldsymbol{y}-\boldsymbol{x}\| \le \epsilon\}$ is a ball of radius ϵ centered at $\boldsymbol{x}$. Then $N(A,\epsilon)$ is the cardinality of the smallest ϵ-net (which is well-defined since if $A \neq \emptyset$ clearly 1 is a lower bound).

Give a counterexample to show that $A \subseteq B$ does *not* imply $N(A,\epsilon) \le N(B,\epsilon)$. *It is true that* $N(A,\epsilon) \le N(B,\epsilon/2)$ *but you don't need to show that; see exercise 4.2.10 in Vershynin.* Your counterexample only needs to show this for one choice of d, once choice of A and B, and once choice of $\epsilon > 0$.

Problem 2: [**APPM 5490 students only**] Like Rademacher complexity, we can define a Gaussian complexity (often referred to as a Gaussian width or Gaussian mean width; definitions often differ by a factor of 2 depending on whether they refer to a radius-like measurement or a diameter-like measurement). For a set $A \subset \mathbb{R}^m$ we'll define

$$\mathcal{G}(A) = \frac{1}{m}\,\mathbb{E}_{\boldsymbol{g}\sim\mathcal{N}(0,\boldsymbol{I}_{m\times m})} \sup_{\boldsymbol{a}\in A} \langle \boldsymbol{g}, \boldsymbol{a}\rangle.$$

a) Show this notion is shift-invariant, i.e., if A is a shifted version of B, then $\mathcal{G}(A) = \mathcal{G}(B)$.

b) If A is the unit Euclidean ball in $\mathbb{R}^m$, calculate $\mathcal{G}(A)$ (or at least a non-trivial bound on it). Your answer should depend on m.

c) If A is the intersection of the Euclidean ball in $\mathbb{R}^m$ with a k-dimensional subspace, calculate $\mathcal{G}(A)$ (or at least a non-trivial bound).

Problem 3: Exercise 3.31 in Mohri et al., slight variant, and different notation. We'll prove a generalization bound directly using covering numbers (as opposed to going through Dudley's inequality). Let $\mathcal{H}$ be a family of functions mapping $\mathcal{X}$ to a subset of real numbers $\mathcal{Y} \subset \mathbb{R}$. In this problem, we consider the covering number with respect to the L^∞ norm on the function space $\mathcal{H}$: $\|h-h'\|_\infty = \max_{x\in\mathcal{X}} |h(x)-h'(x)|$. (We'll assume that the max exists though you could probably just replace it with sup; here, h' just denotes another element in $\mathcal{H}$, *not* the derivative).

Given a dataset $S = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_m)$ with $\boldsymbol{z}_i = (\boldsymbol{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, we'll use the squared loss function $\ell(h, \boldsymbol{z}) = (h(\boldsymbol{x}) - y)^2$ and so we have the generalization error $L_{\mathcal{D}}(h) = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}} \left(h(\boldsymbol{x}) - y\right)^2$ and the empirical risk $\widehat{L}_S(h) = \frac{1}{m}\sum_{i=1}^m \left(h(\boldsymbol{x}_i) - y_i\right)^2$. An advantage of the approach we take is that we can tackle regression (as opposed to classification), but at the expense of having to $\boxed{\text{assume that } \exists M > 0 \text{ such that } |h(\boldsymbol{x}) - y| \leq M}$ for all $h \in \mathcal{H}$ and for all $(\boldsymbol{x}, y) \in \mathcal{X} \times \mathcal{Y}$.

Define the *excess risk* of h to be $E_S(h) = L_{\mathcal{D}}(h) - \widehat{L}_S(h)$ (please don't confuse E with $\mathbb{E}$). Or goal is to bound this for *all* $h \in \mathcal{H}$.

a) Prove that for any S, and any $h, h' \in \mathcal{H}$,

$$|E_S(h) - E_S(h')| \leq 4M\|h - h'\|_\infty.$$

b) Assume $\mathcal{H} \subset \mathcal{B}_1 \cup \mathcal{B}_2 \cup \ldots \cup \mathcal{B}_k$, then show for any $\epsilon > 0$ that

$$\mathbb{P}\left[\sup_{h\in\mathcal{H}} |E_S(h)| \geq \epsilon\right] \leq \sum_{j=1}^k \mathbb{P}\left[\sup_{h\in\mathcal{B}_j} |E_S(h)| \geq \epsilon\right].$$

c) Let $\mathcal{B}_j$ be a ball of radius $\frac{\epsilon}{8M}$ with respect to the L^∞ norm, centered at h_j. Use part (a) to show that

$$\mathbb{P}\left[\sup_{h\in\mathcal{B}_j} |E_S(h)| \geq \epsilon\right] \leq \mathbb{P}\left[|E_S(h_j)| \geq \epsilon/2\right].$$

d) Let $k = N\left(\mathcal{H}, \|\cdot\|_\infty, \frac{\epsilon}{8M}\right)$ be the covering number with respect to the L^∞ norm, and let $\{h_1, \ldots, h_k\} \subset H$ be an ϵ-net with radius $\frac{\epsilon}{8M}$ with corresponding balls $\mathcal{B}_1, \mathcal{B}_2, \ldots \mathcal{B}_k$.

Use Hoeffding's inequality (Theorem D.2 in Mohri et al.) to prove our generalization bound:

$$\mathbb{P}\left[\sup_{h\in\mathcal{H}} |E_S(h)| \geq \epsilon\right] \leq N\left(\mathcal{H}, \|\cdot\|_\infty, \frac{\epsilon}{8M}\right) 2\exp\left(\frac{-m\epsilon^2}{2M^4}\right).$$